LLM Fingerprinting: Which LLM Wrote This Text?

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Code and results: https://github.com/anishhchitnis/llm-identifier

Summary

In this project I trained a CNN and a BiLSTM to guess which LLM family wrote a piece of text. There are five families, GPT, Llama 2, Bard, Falcon and MPT. When the models only see the answer, they get about 0.74 (CNN) and 0.71 (BiLSTM) macro F1, and random guessing would get 0.20. A simple TF IDF and logistic regression baseline did a little better, at 0.77. When the models only see the prompt, they stay close to chance at 0.22. Adding the prompt to the answer did not help. I also tested the models on a domain they never saw in training. The score drops a little for three of the four domains and a lot for Flan. For RQ4 I removed one kind of signal at a time. No single removal broke the classifier, so the signals are spread over word choice, formatting, length and the first and last sentences. The score for the prompt only models is a little above chance, and I found out this comes from a quirk in the data and not from a real fingerprint in the prompt.

1. Problem definition and dataset

1.1 The problem

LLMs write text that sounds a lot alike, so it is hard to tell by reading who wrote it. In this project the model gets a text and has to say which LLM family wrote it. That makes it a 5 class classification problem. I worked on four research questions.

- **RQ1.** Can we identify the LLM family from the output text alone?
- **RQ2.** How does the result change if the model sees only the input (the prompt), only the output, or both?
- **RQ3.** Do the fingerprints carry over to a type of task the model never saw in training?
- **RQ4.** What features of the text tell the LLMs apart, such as length, formatting and vocabulary?

1.2 Where the data comes from

I used the public UltraFeedback dataset from Hugging Face (openbmb/UltraFeedback, MIT license). It has 63,967 prompts. Each prompt has four answers, written by four of 17 different models. Each row has a prompt, an answer, the name of the model that wrote it, and the name of the source dataset the prompt came from. This matches the format the project needs, because each row is (LLM name, input, output). I did not need any API keys, so getting the data was free.

I grouped the models into families. GPT is GPT 4 and GPT 3.5 turbo. Llama 2 is the 7B, 13B and 70B chat models. Bard, Falcon (40B instruct) and MPT (30B chat) each have one model. That gives five families, and the project asks for at least three. I used the source dataset of each prompt as its domain. There are four domains, which are Evol Instruct, Flan, ShareGPT and UltraChat. The Flan source is split into several parts in the data, and I merged them into one domain.

1.3 Cleaning and balancing

- I kept answers between 20 and 8000 characters and dropped duplicates.
- I capped every family and domain pair at 1000 rows, so no single family or domain can dominate. Bard only has 701 Flan answers in the whole dataset, so that one pair has 701 rows.
- Some answers say the name of the model, like "As ChatGPT" or "I am OpenAI". A classifier could just read the name, which is cheating. I found these with a pattern match and removed them. It was about 1 percent of the rows, 183 in total.
- I split the data into train, validation and test with an 80, 10, 10 split. The split is grouped by prompt text, so the same prompt never appears in two splits. I checked that the overlap is zero. I did this because answers to the same prompt look alike, and a random split would let information leak from train to test.

Family	Rows kept	Mean words	Median words	Self naming rows removed
Bard	3,701	268	245	38
Falcon	4,000	103	70	31
GPT	4,000	226	216	52
Llama 2	4,000	267	239	34
MPT	4,000	194	161	28
Total	19,701			183

Table 1. The dataset before the leak rows were removed. Falcon answers are much shorter than the others.

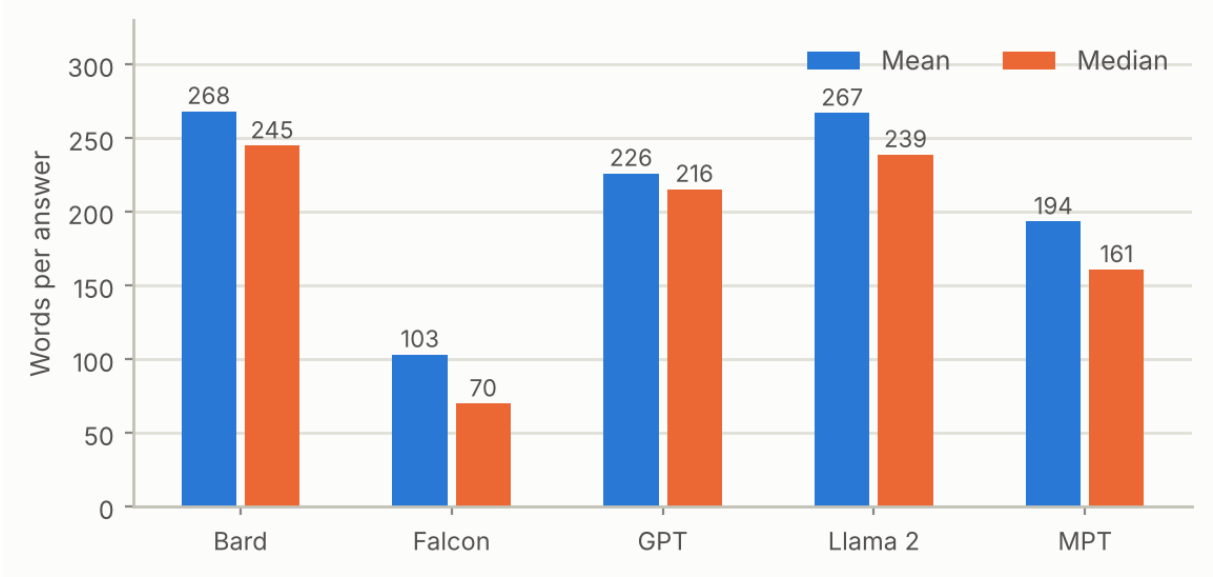


Figure 1. Mean and median answer length for each family. Length is one clear difference between families.

Group	Train	Validation	Test
Bard	2,928	357	378
Falcon	3,206	376	387
GPT	3,071	443	434
Llama 2	3,219	395	352
MPT	3,207	385	380
Evol Instruct	3,937	524	487
Flan	3,767	471	440
ShareGPT	3,920	494	516
UltraChat	4,007	467	488
All rows	15,631	1,956	1,931

Table 2. Rows in each split. All rows is 19,518 after the leak rows were removed.

2. Models and training

2.1 Input to the networks

Both networks use the same tokenizer. It splits text into words, punctuation marks and newlines. It does not lowercase, because capital letters and formatting like bullet points and line breaks can be part of a model's style. The vocabulary has the 30,000 most common tokens from the training set only. Each answer is cut or padded to 384 tokens. For RQ2, the input and output are joined into one sequence. The input is cut to 128 tokens, then a separator token, then the output.

2.2 The CNN

The CNN is a TextCNN. It has a 200 dimension embedding layer, then three parallel 1D convolutions with kernel sizes 3, 4 and 5 and 128 filters each. After each convolution it uses a ReLU and a max pool over the whole text. Padding is masked so it cannot win the max pool. The three pooled vectors are joined, passed through dropout of 0.5, and then a linear layer gives five scores. It has about 6.31 million parameters.

2.3 The RNN

The RNN is a bidirectional LSTM with hidden size 128 in each direction and the same 200 dimension embedding. It uses packed sequences so padding is ignored. I join the mean pool and the max pool of the hidden states, apply dropout of 0.5, and use a linear layer. It has about 6.34 million parameters.

2.4 Training details

Setting	Value
Optimizer	Adam, learning rate 0.001
Batch size	64
Loss	Cross entropy with class weights
Gradient clipping	1.0
Max epochs	20
Early stopping	Stop when validation macro F1 does not improve for 3 epochs, keep the best epoch
Seeds	3 seeds for the CNN and for most BiLSTM results. The BiLSTM runs for RQ3 use 2 seeds. I report mean and standard deviation
Hardware	One T4 GPU on Google Colab

Table 3. Training settings, the same for the CNN and the BiLSTM.

I only used validation data to pick the best epoch and to pick the regularization strength C for the baselines. The test set was used once per final model. Macro F1 is my main number because it treats all five families the same. I also report accuracy.

2.5 Baselines

I also added three simple baselines to compare against. The first is a majority class guess. The second is TF IDF with word 1 and 2 grams plus logistic regression. The third is TF IDF with character 2 to 5 grams plus logistic regression. The character version keeps case, punctuation and newlines. They show how much the neural networks really add.

3. Results

3.1 RQ1, output only

Model	Test accuracy	Test macro F1
Majority class	0.182	0.062
TF IDF words + logistic regression	0.772	0.773
TF IDF characters + logistic regression	0.767	0.769
CNN (3 seeds)	0.737 ± 0.006	0.743 ± 0.005
BiLSTM (3 seeds)	0.714 ± 0.037	0.708 ± 0.042

Table 4. RQ1 results on the test set with 384 tokens of the output. Random guessing among five classes is 0.20 accuracy.

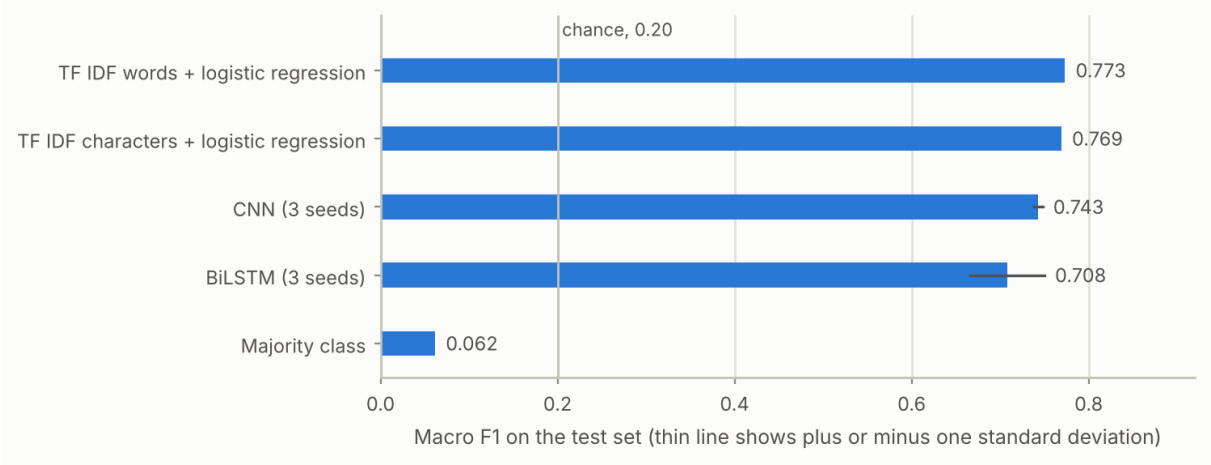


Figure 2. Test macro F1 for each model. The grey line is chance at 0.20.

The answer to RQ1 is yes. All of the real models are far above the 0.20 chance line. The CNN reaches 0.743 macro F1 and the BiLSTM reaches 0.708. The TF IDF baselines are a bit better at 0.773 and 0.769. So on this data the neural networks did not beat a simple linear model on word counts. My guess is that the style signals are strong and local, like the words "Certainly" or "happy to help", and a linear model can use those fine. The CNN is 3.5 points ahead of the BiLSTM, but the BiLSTM changes a lot between seeds, as Table 5 shows, so I would not call that gap firm.

Model	Seed	Test accuracy	Test macro F1	Epochs run
CNN	0	0.744	0.749	8
CNN	1	0.733	0.739	12
CNN	2	0.735	0.740	11
BiLSTM	0	0.679	0.668	11
BiLSTM	1	0.752	0.752	20
BiLSTM	2	0.712	0.703	12

Table 5. Each seed for the output only models at 384 tokens. BiLSTM seed 1 ran all 20 epochs. The other runs stopped early.

3.2 Which families are easy and which are hard

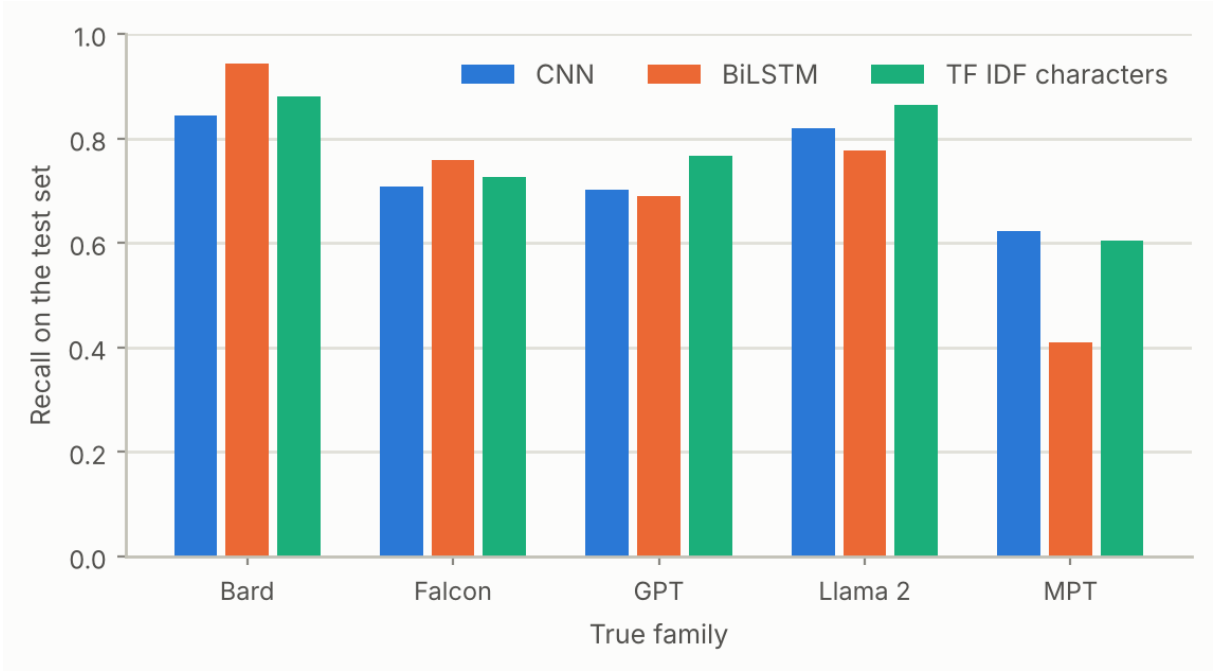


Figure 3. Recall for each true family. Recall is the share of that family's answers that were found.

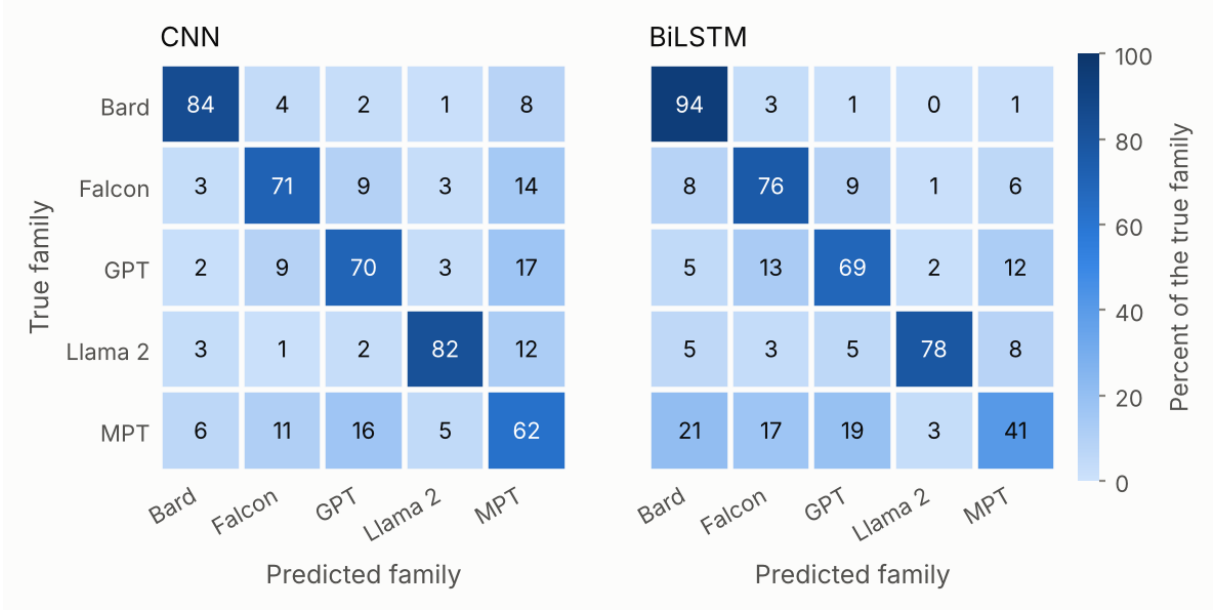


Figure 4. Confusion matrices for the CNN and the BiLSTM. Each row adds to 100 percent. The diagonal is the correct answers.

Bard is the easiest family. The BiLSTM finds 94 percent of Bard answers and the CNN finds 85 percent. Llama 2 is next. MPT is the hardest. The CNN finds 62 percent of MPT answers and the BiLSTM only 41 percent. MPT answers are most often mistaken for GPT and Falcon. GPT and Falcon are also mixed up with each other. The BiLSTM sent many MPT answers to Bard, which is why its MPT recall is so low.

4. Analysis and experiments

4.1 How much text does the model need? (truncation)

I trained both networks again with the output cut to 16, 32, 64, 128, 256 and 384 tokens. Each setting used 3 seeds.

Tokens kept	CNN macro F1	BiLSTM macro F1
16	0.567 ± 0.011	0.561 ± 0.009
32	0.599 ± 0.011	0.587 ± 0.017
64	0.657 ± 0.015	0.633 ± 0.022
128	0.717 ± 0.010	0.707 ± 0.025
256	0.737 ± 0.002	0.727 ± 0.008
384	0.743 ± 0.005	0.708 ± 0.042

Table 6. Test macro F1 when only the first N tokens of the answer are kept.

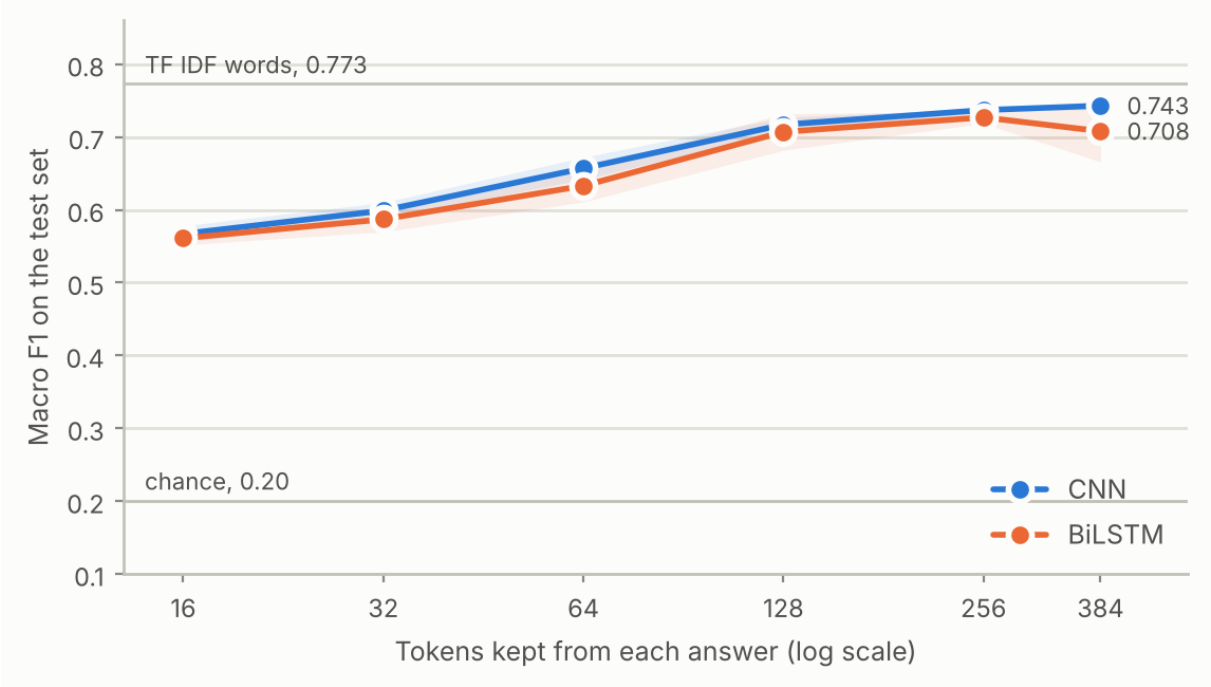


Figure 5. Macro F1 against tokens kept. The shaded band is one standard deviation. The grey line at the top is the best TF IDF baseline.

Even 16 tokens, which is about one short sentence, gives 0.567 for the CNN and 0.561 for the BiLSTM. That is almost three times chance. So the start of an answer already shows a lot of style, like how a model begins with "Sure" or "Certainly". The score rises quickly until about 128 tokens and then flattens. The CNN stays slightly ahead of the BiLSTM at every length, but at 16 tokens they are about the same. The BiLSTM at 384 is lower than at 256, but its spread is large, so I would not read much into that drop.

4.2 RQ2, input only, output only, or both

Model	Output only	Input only	Input + output
CNN	0.743 ± 0.005	0.219 ± 0.004	0.732 ± 0.010
BiLSTM	0.708 ± 0.042	0.217 ± 0.008	0.704 ± 0.007
TF IDF words	0.773	0.226	0.722

Table 7. Test macro F1 for RQ2. Chance is 0.20.

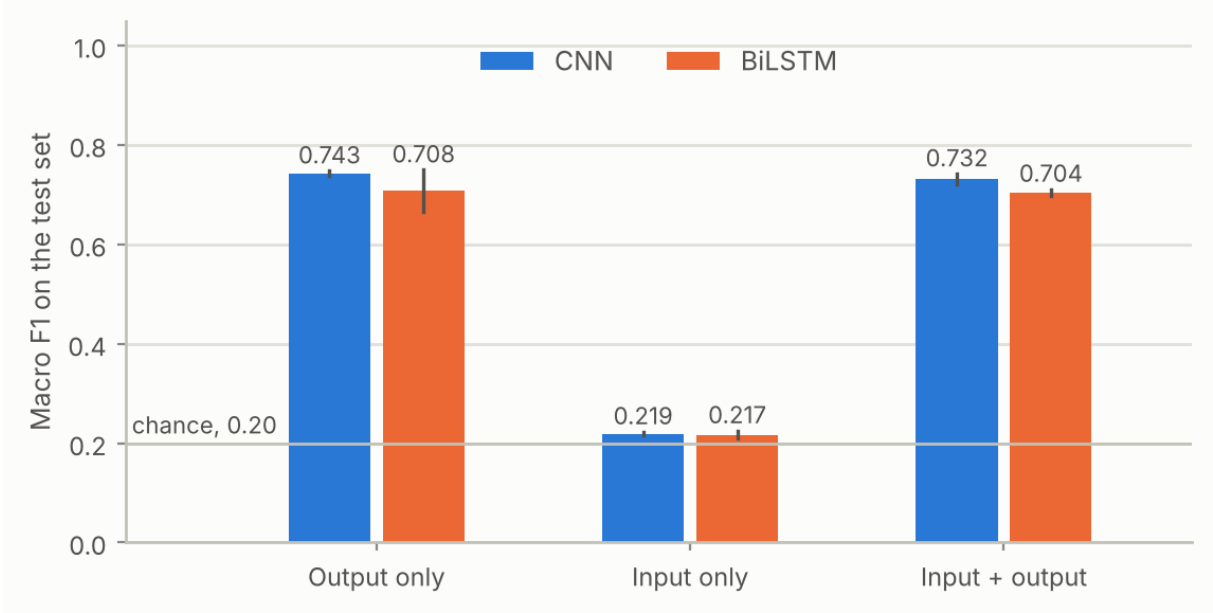


Figure 6. RQ2 results. The grey line is chance at 0.20.

- **Output only** works well, as in RQ1.
- **Input only** is near chance, at 0.219 for the CNN and 0.217 for the BiLSTM. This is what I expected, because the prompt is written before the model answers.
- **Input and output together** is not better than output only. The CNN goes from 0.743 to 0.732 and the BiLSTM from 0.708 to 0.704. The TF IDF model drops more, from 0.773 to 0.722. The prompt adds noise and nothing useful.

So for RQ2, the output has almost all of the signal and the input adds nothing.

4.3 Why is input only a little above chance?

Input only scores 0.22, a little above 0.20. The BiLSTM input only model also finds 55 percent of Bard answers, but only 12 to 22 percent of the answers from each other family. That looked strange, since the prompt should not say who answers. I wanted to know why.

It turned out to be a quirk in the data. Bard has only 701 Flan rows, while every other family has 1000 in each domain. This means that a Flan prompt is a bit more likely to be from a non Bard family overall. But Bard's Flan rows are not spread evenly over prompt lengths. I split the Flan prompts into four equal groups by length and measured the share of Bard rows in each. Bard's overall share of Flan rows is 15 percent, but it is 26 percent for the shortest prompts and only 4 percent for the second shortest.

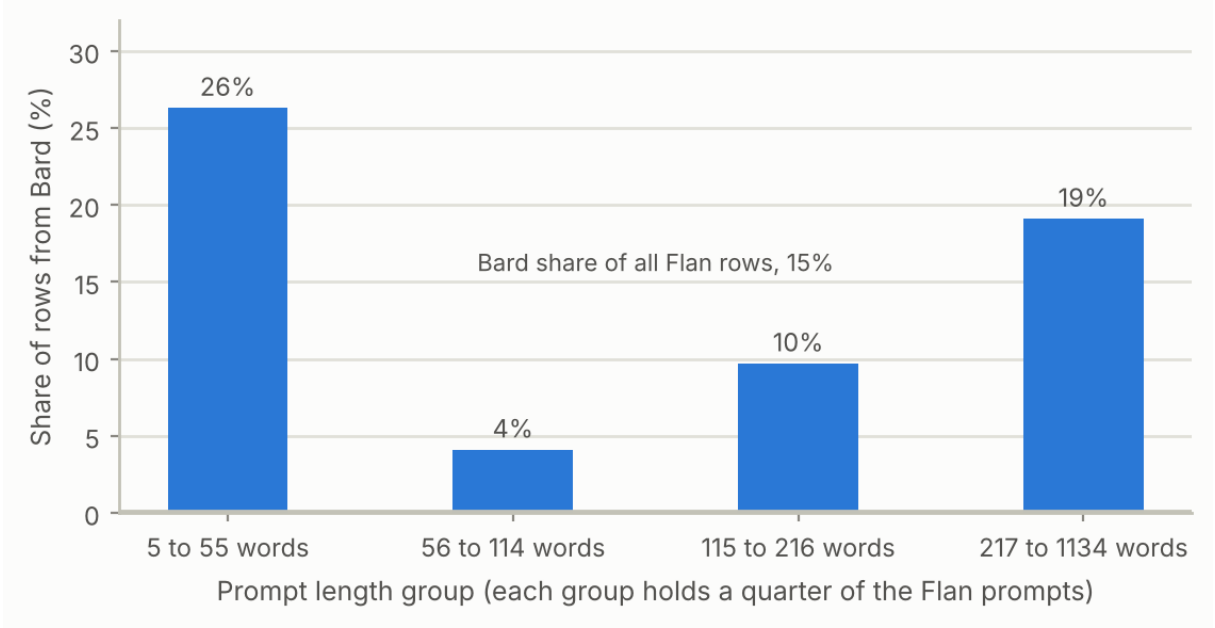


Figure 7. Bard share of Flan rows by prompt length group. If prompt length told us nothing, every bar would be near the 15 percent line.

Domain	Chi square	Degrees of freedom	p value
Evol Instruct	16.9	12	0.153
Flan	286.4	12	below 0.0001
ShareGPT	8.8	12	0.723
UltraChat	6.6	12	0.885

Table 8. Chi square test of prompt length group against family, for each domain. Only Flan shows a link.

Classifier that sees only prompt length and domain	Test accuracy	Test macro F1
Majority class	0.182	0.062
Domain only	0.182	0.133
Prompt length and domain	0.221	0.216

Table 9. A simple classifier that cannot read any words gets the same score as the neural input only models.

The chi square test is strongly significant for Flan and not for the other three domains. A simple classifier that only knows the prompt length and the domain gets 0.216 macro F1. That is the same as the CNN and BiLSTM input only results of 0.219 and 0.217. So the neural models are not finding any fingerprint in the prompt words. They found the Bard and Flan length pattern. This was a shortcut in the data. It shows that checking the data matters as much as building the model.

4.4 What the baselines learned

The top logistic regression features match what I see in real chat output. Llama 2 uses exclamation marks and phrases like "happy to help" and "respectful". Bard uses bullet stars and blank lines. MPT often says "Certainly" and ends with "Overall". GPT uses phrases like "Confidence: 95", which probably come from the way some prompts ask for a confidence level. Falcon features include the words "assistant" and "user" and leftover tag pieces like "< /". Falcon answers are also much shorter, with a median of 70 words against 161 to 245 for the others.

Some of these features say more about the training data and chat template of each model than about the model's deep style. That is still a valid fingerprint, but it may be easy to hide. It also means these results may not carry over to other prompts.

4.5 RQ3, do fingerprints carry over to a new domain?

For this I used the four domains in the data as four types of task. They are Evol Instruct, Flan, ShareGPT and UltraChat. For each domain I trained on the other three and tested on the one I left out. I used the output only models with 384 tokens. To compare fairly, I also scored the normal models (trained on all four domains) on the test rows of that same domain. The CNN used 3 seeds and the BiLSTM used 2 seeds.

Model	Test domain	Seen domain macro F1	Held out macro F1	Drop
CNN	Evol Instruct	0.777 ± 0.010	0.722 ± 0.009	0.055
CNN	Flan	0.707 ± 0.003	0.529 ± 0.016	0.178
CNN	ShareGPT	0.761 ± 0.010	0.687 ± 0.005	0.074
CNN	UltraChat	0.745 ± 0.018	0.704 ± 0.011	0.041
BiLSTM	Evol Instruct	0.702 ± 0.065	0.625 ± 0.010	0.078
BiLSTM	Flan	0.714 ± 0.043	0.513 ± 0.003	0.201
BiLSTM	ShareGPT	0.694 ± 0.060	0.641 ± 0.015	0.052
BiLSTM	UltraChat	0.722 ± 0.063	0.648 ± 0.035	0.074

Table 10. Macro F1 when the test domain was in the training data compared with when it was left out.

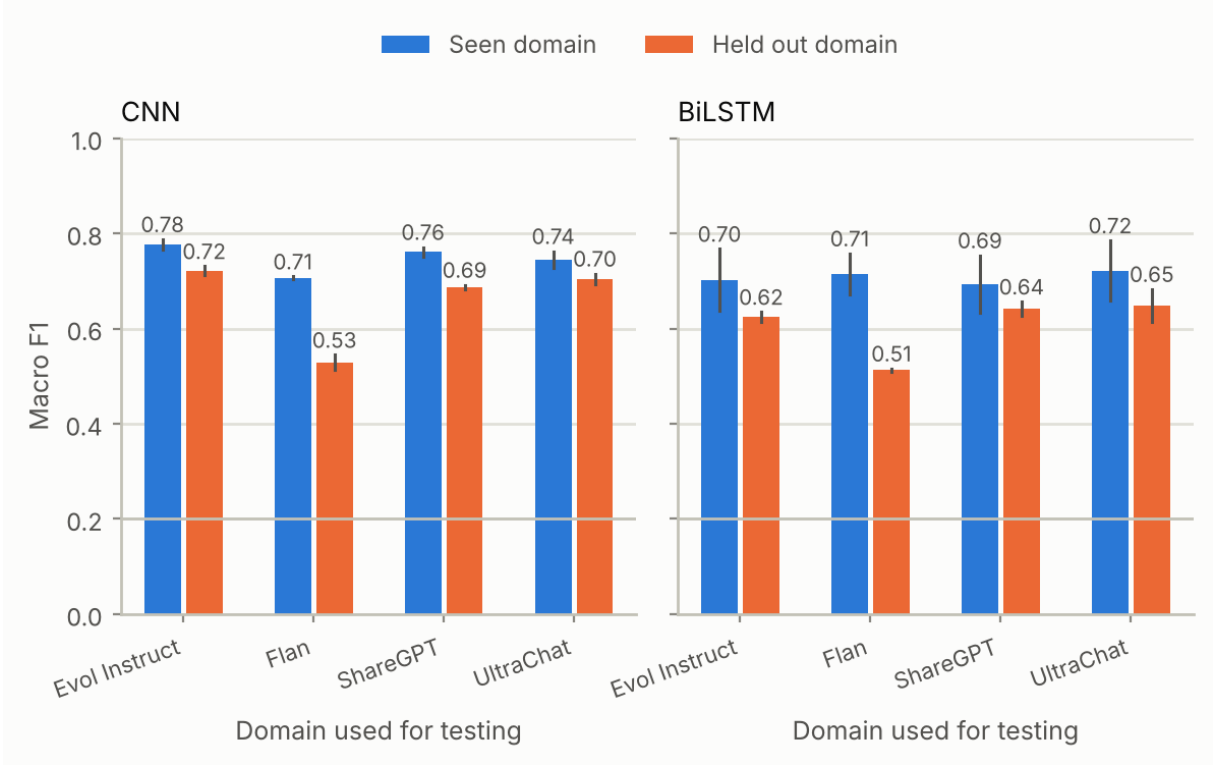


Figure 8. Macro F1 on each domain when it was seen in training (blue) and when it was held out (orange). The grey line is chance at 0.20.

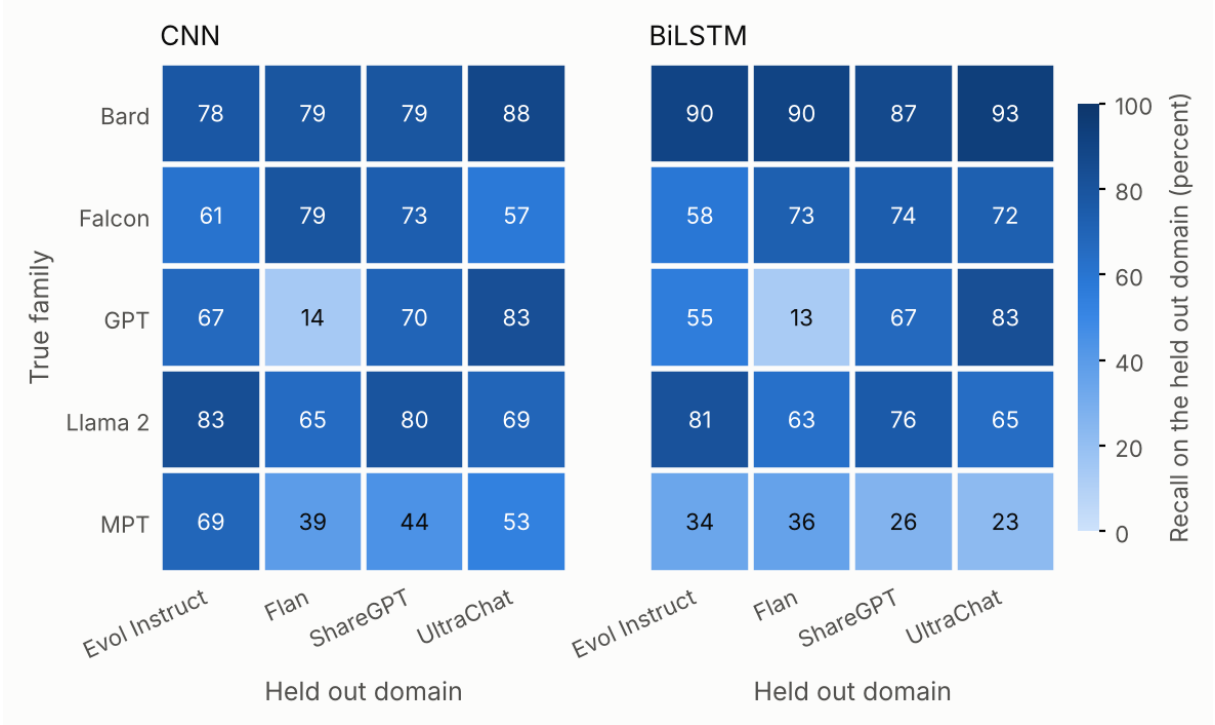


Figure 9. Recall for each family on the held out domain. Each column is a separate model that never saw that domain.

Does performance drop on an unseen domain? Yes, but for three of the four domains the drop is small. For Evol Instruct, ShareGPT and UltraChat the CNN loses 4 to 7 points of macro F1 and the

BiLSTM loses 5 to 8 points. Flan is different. The CNN drops from 0.707 to 0.529 and the BiLSTM from 0.714 to 0.513. That is a loss of about 18 to 20 points. Even then both models are well above the 0.20 chance level. Part of every drop may come from the held out models having less training data, because they train on three domains and not four. So the true drop from the domain change alone is probably a bit smaller than what I show.

Are some families easier to find across domains? Bard is the most stable. Its recall stays between 0.78 and 0.88 for the CNN and between 0.87 and 0.93 for the BiLSTM on every held out domain, even Flan. Llama 2 is also fairly steady, between 0.63 and 0.84. GPT is the least stable. Its recall is 0.55 to 0.83 on the three chat style domains but only 0.14 for the CNN and 0.13 for the BiLSTM on Flan. MPT is weak everywhere, and the BiLSTM only finds 23 to 36 percent of MPT answers in every domain.

Is the classifier learning a general fingerprint or task specific patterns? I think it is both. The scores on the three chat style domains stay close to the seen domain scores, so most of what the models learn is a style that works across tasks. The Flan result shows that some of it is tied to the task. When GPT answers from Flan were tested, the CNN labeled 43 percent of them as Falcon and 37 percent as MPT, and only 14 percent as GPT. Falcon writes short answers, so one possible reason is that the Flan answers are short and the model reads short as Falcon. I did not measure the answer length by domain, so this is a guess and not something I proved.

4.6 RQ4, what tells the LLMs apart?

Accuracy only says that the families can be told apart. Here I look at why. I did two things. First I measured simple features of the answers for each family. Second I removed one kind of signal at a time and checked how much the score fell. These tests use the TF IDF and logistic regression model and a gradient boosted tree model on the features. I did not rerun the CNN and BiLSTM for them.

Feature	Bard	Falcon	GPT	Llama 2	MPT
Words per answer	270	106	229	268	193
Sentences per answer	16.0	5.7	13.2	15.2	10.4
Line breaks per answer	18.5	5.7	19.1	15.2	11.3
Bullet or number lines	3.9	1.1	4.0	3.4	2.6
Bold markers (**) per answer	2.91	0.02	0.34	0.14	0.07
Share of different words in first 100	0.61	0.77	0.72	0.68	0.69
Exclamation marks per word	0.001	0.003	0.002	0.007	0.002
Question marks per word	0.002	0.017	0.003	0.003	0.004
Starts with Sure, Certainly, Of, As or I	0.19	0.15	0.11	0.41	0.15

Table 11. Average of some simple features for each family on the training data.

The families differ in clear ways. Falcon answers are short, with 106 words on average against 270 for Bard and 268 for Llama 2, and they have few line breaks and few bullets. Bard uses far more bold markers, about 2.9 per answer while the others use 0.02 to 0.34. Llama 2 uses the most exclamation

marks, and 41 percent of its answers start with an opener like Sure. Falcon asks the most questions and has the most different words in its first 100 words, which fits its short answers. GPT uses more commas, colons and code marks than the others.

Features used	Macro F1
All structure features	0.641
Only length	0.445
Only layout	0.461
Only punctuation	0.464
Only diversity	0.326
Only voice	0.227
All except length	0.604
All except layout	0.592
All except punctuation	0.589
All except diversity	0.646
All except voice	0.630

Table 12. A tree model that sees only counts like length, layout, punctuation, diversity and voice, never the words themselves.

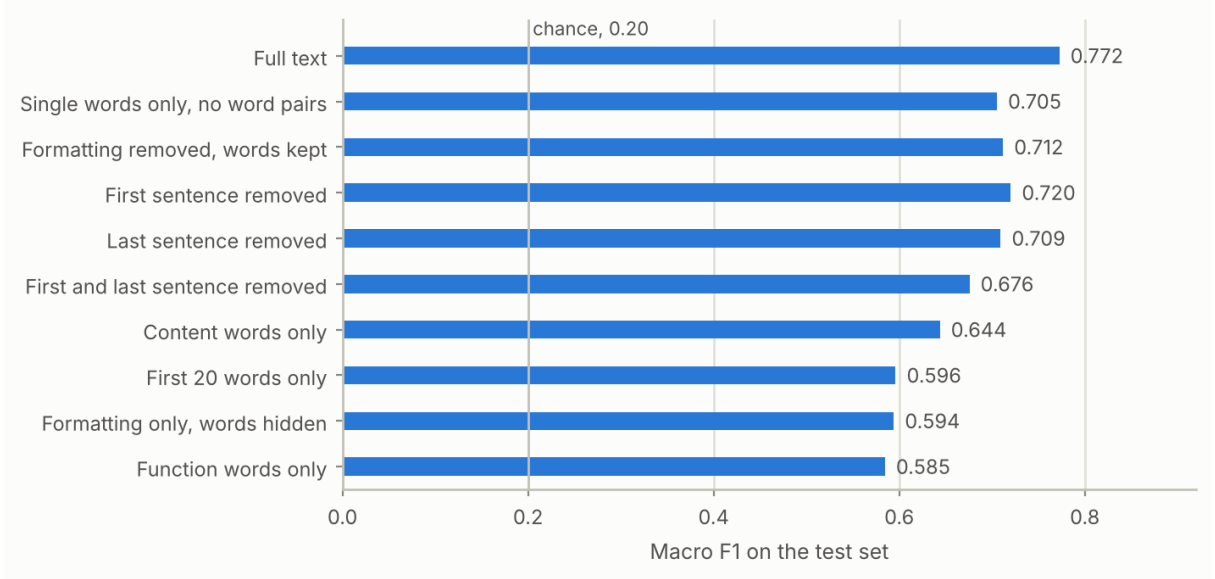


Figure 10. TF IDF and logistic regression after removing or keeping one kind of signal. The top bar is the full text.

What signals does the classifier use, and are they linguistic or structural? Both. A tree model that only sees counts of length, layout, punctuation, diversity and voice gets 0.641 macro F1 without ever seeing a word. The length group alone gets 0.445, layout alone 0.461 and punctuation alone 0.464. Diversity alone gets 0.326 and voice alone only 0.227, so those two are weak. On the word side, keeping the words but removing all formatting still gets 0.712. Keeping only the formatting and hiding

every word gets 0.594. Function words only (like the, to and you) get 0.585 and content words only get 0.644. Each kind of signal carries a lot by itself, and all of them together give the best score of 0.772.

Does removing some signals reduce the performance a lot? Not by itself. Removing the formatting costs 6 points, from 0.772 to 0.712. Removing the word pairs and keeping single words costs 7 points. Removing the first sentence costs 5 points and the last sentence costs 6 points. Removing both the first and last sentence costs 10 points, to 0.676. In the tree model, removing one feature group at a time moves the score between 0.589 and 0.646, and removing the diversity group even helps a little. So no single signal is the whole story. The information is spread out and many of the signals overlap, which is why taking one away does not break the classifier. The first 20 words alone already give 0.596, which matches the truncation result in 4.1.

5. Limits of this work

- All text comes from one dataset, UltraFeedback. The results may not hold for other prompts or for newer models.
- A family can have more than one model, such as GPT 4 and GPT 3.5, and Llama 2 in three sizes. I did not test whether the classifiers could tell models in the same family apart.
- In UltraFeedback each prompt gets four of the 17 models, so not every family answered every prompt.
- Bard has fewer rows than the other families because of how much Bard data exists in the source.
- I used 3 seeds, so the standard deviations are rough. I did not do a heavy search for hyperparameters.
- The RQ4 tests use a linear model and a tree model, not the CNN and BiLSTM, and each setting was run once with a fixed regularization strength. They show which signals carry information, but they do not prove that the neural networks use the same signals.
- For RQ3, the held out models trained on about three quarters of the data, and the BiLSTM used only 2 seeds. The four domains are also not fully separate tasks. They are different sources of prompts, and some of them mix several kinds of tasks.

6. Lessons and experience

The biggest thing I learned is that the data took more work than the models. At the start I thought the CNN and the LSTM would be the hard part, but they were short to write and trained fine. Getting the data right took a lot longer. My first dataset only had three domains, because the Flan prompts had a different name than I expected and I missed them. I also had to find the answers where a model says its own name, because a classifier could just read the name and look smart for the wrong reason.

I also learned why you split by prompt. A random split looks fine, but then the same prompt can be in train and in test with different answers, and the scores come out too high. Grouping by prompt fixed that.

It surprised me that the simple TF IDF baseline beat my neural networks. I was a bit disappointed at first, but it is a useful result. It tells me the style signal is mostly in which words and symbols show up, and not in long range structure. If I had skipped the baseline I would have thought 0.74 was great without knowing that a much simpler model gets 0.77.

The input only result taught me the most. A score of 0.22 is tiny and I almost called it noise. But it showed up in both models, so I looked into it. The reason was that Bard has fewer Flan rows. It would have been easy to say the prompt leaks a little about the model, and that would have been wrong. Next time I will check the simple explanations before I make a claim.

The held out domain test also taught me something. A model that scores 0.74 on average can fall to 0.53 on one domain, so one average score hides a lot. Something I would like to try next is testing the models on a family that was not in training, and looking at the wrong predictions one by one to see what the MPT answers that look like GPT have in common.

Reproducing this

All code is in the repo under src. The scripts run in this order, build_dataset.py, preprocess.py, baselines.py, train.py for the CNN and BiLSTM runs, and rq2_diagnostics.py. The neural runs need a GPU. I used a Colab T4. The file make_figures.py draws the figures from the files in the results folder.